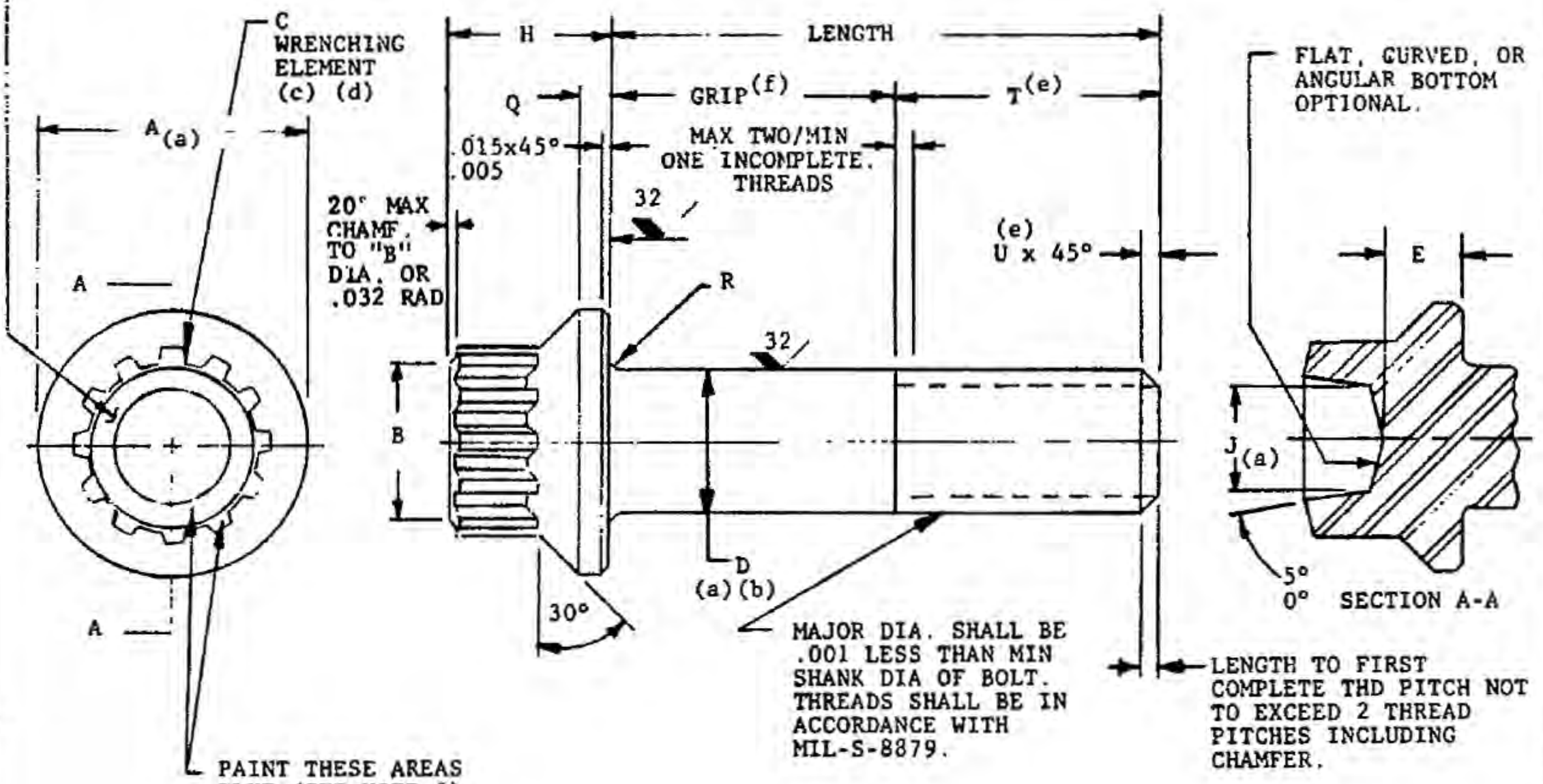
FED-SUP CLASS
5306

STAMP MS14163, DIA. DASH NO.
GRIP DASH NO. AND MANUFACTURER'S
IDENTIFICATION, DEPRESSED .010 MAX



NOTES:

1. MATERIAL: ALLOY STEEL, AMS 6487
2. HARDNESS: ROCKWELL "C" 46-50.
3. SURFACE TEXTURE IN ACCORDANCE WITH ANSI B46.1.1971.
4. PLATING: CADMIUM COATED (VACUUM DEPOSITED): MIL-C-8837, TYPE II, CLASS 2.
5. PAINT: AFTER BAKE, TOP OF BOLT HEAD WITH NO. 15102 BLUE PER TT-E-489.
6. DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES: DECIMALS ±.010, ANGLES ±2°.
7. DIMENSIONS TO BE MET AFTER PLATING.
8. DESIGN AND USAGE INFORMATION: THESE BOLTS ARE DESIGNED TO BE USED WITH MS14164 NUTS AND MS14177 WASHERS.
9. PERFORMANCE: PER MIL-B-8906 EXCEPT FOR:
   (a) FATIGUE PER MIL-B-8906 EXCEPT THAT LOADING SHALL BE AS SHOWN IN MS14163 TABLE II, AND MS14164 NUTS, MS WASHERS SHALL BE USED, NO BOLT SHALL FAIL AT LESS THAN 2,000,000 CYCLES.
   THREADS AND BEARING SURFACES OF MS14163 BOLTS, MS14164 NUTS AND MS WASHERS SHALL BE LUBRICATED WITH MIL-T-5544 COMPOUND FOR THIS TEST.

APPROVED 3 DEC 75 REVISED (A) 30 JUNE 1977 (B) 15 AUGUST 1978

FOR DESIGN FEATURE PURPOSES, THIS STANDARD TAKES PRECEDENCE OVER PROCUREMENT DOCUMENTS REFERENCED HEREIN. REFERENCED DOCUMENTS SHALL BE OF THE ISSUE IN EFFECT ON DATE OF INVITATIONS FOR BID.

(R) REVISED AND REDRAWN

P.A. NAVY - AS Other Cust USAF - 11 ARMY - AV	TITLE BOLT, WHEEL, TENSION, FLANGED, STEEL 220 KSI, Ftu, 450°F. EXTERNAL WRENCHING, SPLINE, DRIVE	MILITARY STANDARD MS14163
PROCUREMENT SPECIFICATION MIL-B-8906	SUPERSEDES	SHEET 1 OF 3

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PLATE NO. 13965

FED. SUP CLASS
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TABLE I

DIA. DASH NO.	NOMINAL SIZE	THREAD UNJF-3A (b)	A DIA. (a)		B DIA.		C (c) (d) WRENCHING ELEMENT	D DIA. (a) (b)		E MIN	H ±.010	J (a) (c) +.010 -.030
			MAX	MIN	MAX	MIN		MAX	MIN			
04	1/4	.2500-28	.438	.428	.250	.235	10	.2495	.2485	.149	.250	.125
05*	5/16	.3125-24	.531	.521	.375	.360	12	.3120	.3110	.187	.312	.215
06	3/8	.3750-24	.640	.639	.438	.423	14	.3745	.3735	.224	.337	.260
07	7/16	.4375-20	.750	.740	.500	.485	16	.4370	.4360	.262	.393	.320
08	1/2	.5000-20	.828	.818	.562	.547	20	.4995	.4985	.299	.450	.380
09	9/16	.5625-18	.938	.928	.625	.610	22	.5615	.5605	.336	.505	.440
10	5/8	.6250-18	1.050	1.040	.688	.673	24	.6240	.6230	.374	.562	.500
12	3/4	.7500-16	1.230	1.220	.812	.797	28	.7490	.7480	.449	.600	.570
14	7/8	.8750-14	1.438	1.428	.938	.923	32	.8740	.8730	.523	.700	.650

TABLE II

DIA. DASH NO.	M ±.010	Q ±.010	R RAD		T MIN REF (e)	U REF (e)	FATIGUE LOADING		ULTIMATE TENSILE STRENGTH LB MIN	DOUBLE SHEAR STRENGTH LB MIN	CONCENTRICITY		
			MAX	MIN			LOW (g) TENSION LOAD LB ±2%	HIGH (h) TENSION LOAD LB ±2%			X (a)	Y (c)	Z (b)
04	.062	.045	.041	.031	.560	.045	5,800	6,850	8,890	13,000	.005	.004	.0045
05		.061			.663	.052	9,200	10,900	14,100	20,200	.006	.005	
06	.070	.069			.733		13,600	16,100	20,900	29,100	.008	.006	
07		.082	.057	.047	.823	.063	18,400	21,800	28,300	39,700	.009	.008	.0060
08		.102			.903		24,600	29,100	37,800	51,800	.010	.008	
09	.094	.112			1.000	.069	31,100	34,500	47,900	65,600	.011	.009	
10		.123			1.060		38,900	43,100	59,900	81,000	.012	.010	
12		.155	.073	.063	1.241	.078	56,500	62,600	86,900	117,000	.015	.012	
14		.180			1.427	.089	77,400	85,700	119,000	159,000	.018	.014	.0090

NOTES:
(a) DIMENSION A, D AND J SHALL BE CONCENTRIC TO EACH OTHER WITHIN VALUES SPECIFIED FOR X T.I.R.
(b) DIMENSIONS D AND THREAD PITCH DIAMETER SHALL BE CONCENTRIC TO EACH OTHER WITHIN VALUES SPECIFIED FOR Z T.I.R.
(c) DIMENSIONS J AND WRENCHING ELEMENT SHALL BE CONCENTRIC TO EACH OTHER WITHIN VALUES SPECIFIED FOR Y T.I.R.
(d) WRENCHING ELEMENT PER MS33787. DRIVERS PER MIL-W-8982.
(e) REFERENCE DIMENSIONS ARE FOR DESIGN PURPOSES ONLY AND ARE NOT INSPECTION OR MANUFACTURING REQUIREMENTS. T = H MAX (MS 14156) + 3 THDS + .153
(f) GRIP LENGTH OF BOLT SHALL BE MEASURED FROM UNDERSIDE OF HEAD TO THE END OF FULL CYLINDRICAL PORTION OF SHANK.
(g) LOW TENSION FATIGUE LOAD IS BASED ON .65 x Ftu OF TABLE II.
(h) HIGH TENSION FATIGUE LOAD FOR SIZES 1/4 THROUGH 1/2 INCH ARE BASED ON .77 x Ftu AND SIZES 9/16 THROUGH 7/8 INCH ARE BASED ON .72 x Ftu OF TABLE II.

REVISED
APPROVED 3 DEC 75
FOR CHANGES SEE SHEET 1

P.A. NAVY - AS Other Cust USAF - 11 ARMY - AV	TITLE BOLT, WHEEL, TENSION, FLANGED, STEEL, 220 KSI, Ftu, 450°F EXTERNAL WRENCHING, SPLINE DRIVE	MILITARY STANDARD MS14163
PROCUREMENT SPECIFICATION MIL-B-8906	SUPERSEDES	SHEET 2 OF 3

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TABLE III

GRIP DASH NO.	GRIP ± .010	DIAMETER DASH NUMBER								
		04	05	06	07	08	09	10	12	14
		LENGTH ± .015								
004	.250	.835	.938	—	—	—	—	—	—	—
006	.375	.960	1.063	1.133	1.223	—	—	—	—	—
008	.500	1.085	1.188	1.258	1.348	1.428	—	—	—	—
010	.625	1.210	1.313	1.383	1.473	1.553	1.650	1.710	—	—
012	.750	1.335	1.438	1.508	1.598	1.678	1.775	1.835	2.016	—
014	.875	1.460	1.563	1.633	1.723	1.803	1.900	1.960	2.141	2.327
016	1.000	1.585	1.688	1.758	1.848	1.928	2.025	2.085	2.266	2.452
018	1.125	1.710	1.813	1.883	1.973	2.053	2.150	2.210	2.391	2.577
020	1.250	1.835	1.938	2.008	2.098	2.178	2.275	2.335	2.516	2.702
022	1.375	1.960	2.063	2.133	2.223	2.303	2.400	2.460	2.641	2.827
024	1.500	2.085	2.188	2.258	2.348	2.428	2.525	2.585	2.766	2.952
026	1.625	2.210	2.313	2.383	2.473	2.553	2.650	2.710	2.891	3.077
028	1.750	2.335	2.438	2.508	2.598	2.678	2.775	2.835	3.016	3.202
030	1.875	2.460	2.563	2.633	2.723	2.803	2.900	2.960	3.141	3.327
032	2.000	2.585	2.688	2.758	2.848	2.928	3.025	3.085	3.266	3.452
034	2.125	2.710	2.813	2.883	2.973	3.053	3.150	3.210	3.391	3.577
036	2.250	2.835	2.938	3.008	3.098	3.178	3.275	3.335	3.516	3.702
038	2.375	2.960	3.063	3.133	3.223	3.303	3.400	3.460	3.641	3.827
040	2.500	3.085	3.188	3.258	3.348	3.428	3.525	3.585	3.766	3.952
042	2.625	3.210	3.313	3.383	3.473	3.553	3.650	3.710	3.891	4.077
044	2.750	3.335	3.438	3.508	3.598	3.678	3.775	3.835	4.016	4.202
046	2.875	3.460	3.563	3.633	3.723	3.803	3.900	3.960	4.141	4.327
048	3.000	3.585	3.688	3.758	3.848	3.928	4.025	4.085	4.266	4.452
050	3.125	3.710	3.813	3.883	3.973	4.053	4.150	4.210	4.391	4.577
052	3.250	3.835	3.938	4.008	4.098	4.178	4.275	4.335	4.516	4.702
054	3.375	3.960	4.063	4.133	4.223	4.303	4.400	4.460	4.641	4.827
056	3.500	4.085	4.188	4.258	4.348	4.428	4.525	4.585	4.766	4.952
058	3.625	4.210	4.313	4.383	4.473	4.553	4.650	4.710	4.891	5.077
060	3.750	4.335	4.438	4.508	4.598	4.678	4.775	4.835	5.016	5.202
062	3.875	4.460	4.563	4.633	4.723	4.803	4.900	4.960	5.141	5.327
064	4.000	4.585	4.688	4.758	4.848	4.928	5.025	5.085	5.266	5.452
066	4.125	4.710	4.813	4.883	4.973	5.053	5.150	5.210	5.391	5.577
068	4.250	4.835	4.938	5.008	5.098	5.178	5.275	5.335	5.516	5.702
070	4.375	4.960	5.063	5.133	5.223	5.303	5.400	5.460	5.641	5.827
072	4.500	5.085	5.188	5.258	5.348	5.428	5.525	5.585	5.766	5.952
074	4.625	5.210	5.313	5.383	5.473	5.553	5.650	5.710	5.891	6.077
076	4.750	5.335	5.438	5.508	5.598	5.678	5.775	5.835	6.016	6.202
078	4.875	5.460	5.563	5.633	5.723	5.803	5.900	5.960	6.141	6.327
080	5.000	5.558	5.688	5.758	5.848	5.928	6.025	6.085	6.266	6.452
082	5.125	5.710	5.813	5.883	5.973	6.053	6.150	6.210	6.391	6.577
084	5.250	5.835	5.938	6.008	6.098	6.178	6.275	6.335	6.516	6.702
086	5.375	5.960	6.063	6.133	6.223	6.303	6.400	6.460	6.641	6.827
088	5.500	6.085	6.188	6.258	6.348	6.428	6.525	6.585	6.766	6.952
090	5.625	6.210	6.313	6.383	6.473	6.553	6.650	6.710	6.891	7.077
092	5.750	6.335	6.438	6.508	6.598	6.678	6.775	6.835	7.016	7.202
094	5.875	6.460	6.563	6.633	6.723	6.803	6.900	6.960	7.141	7.327
096	6.000	6.585	6.688	6.758	6.848	6.928	7.025	7.085	7.266	7.452
098	6.125	6.710	6.813	6.883	6.973	7.053	7.150	7.210	7.391	7.577
100	6.250	6.835	6.938	7.008	7.098	7.178	7.275	7.335	7.516	7.702
102	6.375	6.960	7.063	7.133	7.223	7.303	7.400	7.460	7.641	7.827
104	6.500	7.085	7.188	7.258	7.348	7.428	7.525	7.585	7.766	7.952
106	6.625	7.210	7.313	7.383	7.473	7.553	7.650	7.710	7.891	8.077
108	6.750	7.335	7.438	7.508	7.598	7.678	7.775	7.835	8.016	8.202
110	6.875	7.460	7.563	7.633	7.723	7.803	7.900	7.960	8.141	8.327
112	7.000	7.585	7.688	7.758	7.848	7.928	8.025	8.085	8.266	8.452
114	7.125	7.710	7.813	7.883	7.973	8.053	8.150	8.210	8.391	8.577
116	7.250	7.835	7.938	8.008	8.098	8.178	8.275	8.335	8.516	8.702
118	7.375	7.960	8.063	8.133	8.223	8.303	8.400	8.460	8.641	8.827
120	7.500	8.085	8.188	8.258	8.348	8.428	8.525	8.585	8.766	8.952
122	7.625	8.210	8.313	8.383	8.473	8.553	8.650	8.710	8.891	9.077
124	7.750	8.335	8.438	8.508	8.598	8.678	8.775	8.835	9.016	9.202
126	7.875	8.460	8.563	8.633	8.723	8.803	8.900	8.960	9.141	9.327

REVISED (R) FOR CHANGES SEE SHEET 1
APPROVED 3 DEC 75

DIAMETER DASH NO. AT TOP OF TABLE, PLUS GRIP DASH NO. CONSTITUTES A COMPLETE PART NO.

FIRST TWO DASH NUMBERS DESIGNATE DIAMETER IN SIXTEENTHS. LAST THREE DASH NUMBERS DESIGNATE GRIP LENGTH IN SIXTEENTHS.

UNASSIGNED DASH NUMBERS SHALL NOT BE USED.

EXAMPLE OF PART NUMBER: MS14163-05012

- MS14163 — BASIC NUMBER
- 05012 — 5/16-24 BOLT, .750 GRIP, 1.438 LENGTH

P.A.
NAVY - AS
Other Cust
USAF - 11
ARMY - AV

TITLE
BOLT, WHEEL, TENSION, FLANGED, STEEL, 220 KSI, Ftu, 450°F, EXTERNAL WRENCHING, SPLINE DRIVE

MILITARY STANDARD
MS14163

PROCUREMENT SPECIFICATION
MIL-B-8906

SUPERSEDES

SHEET 3 OF 3

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